

AUTOMATIC WASTE SORTER & DISPOSAL SYSTEM

Design, Hardware Implementation, and Multi-Sensor Waste Classification Report

PROJECT TYPE	CONTROL SYSTEM	ACTUATION	STATUS
Embedded Automation	Microcontroller Board	Stepper & ULN2003	Functional Prototype

Executive Summary

This report documents the design and functional testing of an automated waste segregation and disposal system. The prototype uses a multi-sensor array—combining an IR proximity sensor, an inductive proximity sensor, and a rain/moisture sensor—to automatically classify waste into metallic, wet, and dry/plastic categories. Upon detection, an audio buzzer alert is triggered, and a stepper motor coupled with a ULN driver indexes a rotary base to position the correct disposal bin beneath the drop mechanism.

1. SYSTEM ARCHITECTURE & COMPONENTS

The prototype integrates several hardware components organized into three primary subsystems: Sensing, Processing/Control, and Actuation/Feedback.

Sensing Subsystem IR Sensor: Detects presence of waste item in entry chute. Inductive Sensor: Detects metallic materials. Rain Sensor: Measures moisture content for wet waste classification.	Control Unit Central microcontroller processing sensor inputs in real-time, executing logic routines to classify material type and driving output signals for motor positioning and feedback.	Actuation & Interface ULN Driver & Stepper: Rotates the multi-bin base precisely. Drop Servo Flap: Releases waste into target bin. Piezo Buzzer: Audio feedback.
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2. MATERIAL DETECTION LOGIC MATRIX

The decision tree evaluates inputs sequentially to determine bin allocation:

IR Sensor (Presence)	Inductive Sensor	Rain / Moisture Sensor	Assigned Bin Category	Target Bin Position
HIGH (Detected)	HIGH (Metallic)	DON'T CARE	Metallic Waste	Bin 1 (Metal Collection)
HIGH (Detected)	LOW (Non-metal)	HIGH (Moisture > Threshold)	Wet Waste	Bin 2 (Organic / Wet)
HIGH (Detected)	LOW (Non-metal)	LOW (Dry)	Dry / Plastic Waste	Bin 3 (Plastic & Paper)
LOW (Empty)	LOW	LOW	Standby / Idle	Home Position

3. HARDWARE SPECIFICATIONS & INTERFACING

The physical rig is constructed on an integrated mounting plate featuring a vertical support frame, a primary detection column with drop-flap, and a rotating platter holding three clear waste containers.

Component Name	Specifications / Model	Role in System
Stepper Motor	28BYJ-48 (5V DC) / NEMA 17	Rotates the bin selector turntable to angular target.
Motor Driver	ULN2003 / ULN Driver Board	Amplifies control signals from microcontroller to drive motor coils.
Inductive Sensor	LJ12A3-4-Z/BX (6-36V)	Identifies ferrous and conductive metallic waste items.
Rain / Moisture Sensor	FC-37 / YL-83 Board	Measures conductive liquid contact for wet waste detection.
IR Sensor Module	TCRT5000 / Standard IR Module	Triggers the detection cycle upon item insertion.
Buzzer & Indicator	5V Active Piezo Buzzer & LEDs	Provides instant audio/visual confirmation upon detection.

4. OPERATIONAL WORKFLOW & TEST RESULTS

System operation follows a structured loop executed during testing and demonstrated in operational verification:

- Idle State:** System monitors IR sensor while turntable rests at home position.
- Object Insertion:** Object is introduced into upper chute. IR sensor transitions HIGH.
- Audio Notification:** Piezo buzzer sounds a distinct beep signaling waste detection.
- Sensor Sampling:** Controller reads Inductive and Moisture lines within a 500ms window.
- Positioning:** Stepper motor energized via ULN driver rotates the base plate to align the selected bin (0°, 120°, or 240°).
- Disposal & Reset:** Trap door flap releases object into designated bin; turntable resets to default state.

5. KEY FINDINGS & FUTURE ENHANCEMENTS

The prototype successfully demonstrated automated classification across metal, wet, and plastic/dry samples. Future iterations can incorporate weight sensors (load cells) and image-recognition camera modules for higher precision sorting.